COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	9/9/2026	**DATE SUBMITTED:**	9/9/2026
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ACTIVITY NO. 8 SORTING ALGORITHMS

1. PROCEDURE OUTPUT

Code

```cpp
const int maxSize = 100;

void displayArray(int arr[]);

int main(){
    int array[maxSize];
    srand(time(0));
    for(int i = 0; i < maxSize; i++){
        array[i]= rand() % 100;
    }
    displayArray(array);
}

void displayArray(int arr[]){
    for(int i = 0; i < maxSize; i++){
        std::cout << arr[i] << " ";
    }
}
```

Console:

```
94 5 29 12 61 1 66 82 12 35 46 69 65 51 91 34 86 52 24 84 17 98 48 0 98 39 8 25 34 68 21 35 77 86 19 17 75 73 83 95 55 13 26 92 96 47 58 88 35 47 60
 78 15 28 98 56 91 93 43 30 17 87 76 19 2 86 58 1 64 77 54 26 46 56 62 8 54 53 49 9 54 95 7 46 13 24 92 84 26 13 58 68
```

Observations:
This shows the random values that are generated inside of an array for sorting algorithm testing..

Table 8-1. Array of Values for Sort Algorithm Testing

Code:

Console:
Observations:

Table 8-2. Shell Sort Technique

Code:
Console:
Observations:

Table 8-3. Merge Sort Algorithm

Code:
Console:
Observations:

Table 8-4. Quick Sort Algorithm

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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